

DATA ANALYSIS INTERNSHIP

Healthcare No-Shows

Why Patients Miss Their Appointments:
Data Cleaning, KPIs and an Interactive Excel Dashboard

Data • Data Quality • Cleaning • KPIs
Results • Insights • Excel Dashboard

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1 Executive summary

The task is to turn the raw file `healthcare_noshows.csv` into a professional, interactive Excel dashboard: clean and prepare the data, define the relevant KPIs, build the charts and extract the insights that explain why patients miss their appointments. The whole work is done in Excel, in one workbook: `Healthcare_NoShows_Dashboard.xlsx`.

The file holds **106,987 appointments** held between 29 April and 8 June 2016. After cleaning, **106,977 appointments** of **60,268 patients** remain, and **21,672** of them were missed: **one appointment in five (20.3%)**.

Waiting is the main driver. An appointment made for the same day is missed in **4.7%** of cases; an appointment booked one day or more ahead in **28.5%**, six times more.

The SMS reminder works, although the raw figures say the opposite. Patients who received an SMS miss more often (27.7% against 16.7%), because the reminder is only sent for appointments booked several days ahead. Compared at the same lead time, the SMS lowers the no-show rate in every band; the Mantel-Haenszel odds ratio computed in the workbook is **0.78**.

Past behaviour predicts the next visit. After a previous no-show the rate is **29.4%**, against 17.1% after an attended visit. Teenagers (26.4%) and welfare recipients (+3.9 points) also miss more; the weekday makes almost no difference.

The file supports a simple targeting rule: remind every patient booked one day or more ahead, and call the patients who already missed an appointment.

2 Data and method

2.1 The data

The file supplied with the task has 15 columns and one row per appointment: the patient and appointment identifiers (`PatientId`, `AppointmentID`), the gender and age, the neighbourhood of the health unit, the booking and appointment dates (`ScheduledDay`, `AppointmentDay`) and their difference in days (`Date.diff`), five health and social flags (`Scholarship`, `Hipertension`, `Diabetes`, `Alcoholism`, `Handcap`), the reminder flag `SMS_received` and the outcome `Showed_up`.

2.2 Method

The work follows the brief step by step, with Excel tools only.

1. **Audit:** completeness, duplicates, ranges and chronology of the dates (Excel table, COUNTIFS, COUNTBLANK).

2. **Cleaning:** invalid rows removed, columns renamed, codes recoded (recoding tables, PROPER, SUBSTITUTE).
3. **Preparation:** nine analysis fields added by formula (IF, CHOOSE, WEEKDAY).
4. **KPIs:** nine indicators, each answering one question (AVERAGE, SUM, AVERAGEIFS).
5. **Analysis:** rates by segment and chi-square tests of the differences (COUNTIFS, SUMIFS, CHISQ.DIST.RT).
6. **Dashboard and insights:** two interactive dashboards (pivot tables, pivot charts, slicers, drop-down lists) and seven findings, each with an action and an owner.

Every number of this report is read from the workbook, where it is a live formula on the clean table `tblAppointments`.

3 Data quality

Table 1: Audit of the raw file (106,987 rows)

Check	Rows	Decision
Empty cells	0	None found
Duplicate AppointmentID	0	None found
Date.diff different from the two dates	0	None found
Appointment dated before its booking	5	Removed: impossible chronology
Age of 115 years	5	Removed: implausible age
PatientId with decimals	5	Kept, stored as text
Second booking of a patient on the same day	8,532	Kept: own identifier and outcome
Appointments on a Saturday	39	Kept: rare but valid
Misspelled column names	3	Renamed
Codes TRUE/FALSE, F/M, upper-case names	106,987	Recoded

The file is complete and internally consistent: no cell is empty, no appointment is duplicated, and `Date.diff` always equals the difference between the two dates. Two defects make a row unusable. Five appointments are dated before their booking, which is impossible; five others belong to two patients recorded at 115 years, an implausible age. The other findings are valid situations (several bookings on one day, rare Saturday clinics) or presentation issues.

4 Cleaning and preparation

4.1 Cleaning

The ten invalid rows are removed and listed in full on the *Data Cleaning* sheet, so that the removal can be checked. Three columns are renamed (Hipertension to Hypertension, Handcap to Handicap, Date.diff to LeadTimeDays). TRUE/FALSE become Yes/No, F/M become Female/Male, and `Showed_up` becomes `Status` (Attended, No-show). The neighbourhood names, written in capitals, are converted with the formula `PROPER`, the particles da, de, do and das being set back in lower case with `SUBSTITUTE`; the 81 names and their formula are shown on the same sheet.

4.2 Preparation

Nine fields are added to the Excel table `tblAppointments`, each by a formula copied down the column, so that the table stays consistent if a value is corrected.

Table 2: Fields added by Excel formulas

Field	Definition
NoShow	1 for a no-show, 0 otherwise: its average is directly the no-show rate
LeadTimeDays	AppointmentDate minus ScheduledDate
LeadTimeBand	Same day, 1 to 3, 4 to 7, 8 to 14, 15 to 30, 31 to 60, 61+ days (nested IF)
AgeGroup	00-11, 12-17, 18-24, 25-34, 35-44, 45-54, 55-64, 65-74, 75+ (nested IF)
Weekday, WeekStart	Day name (CHOOSE and WEEKDAY) and Monday of the appointment week
SMSFlag	1 if an SMS was received: its average is the SMS coverage
PriorNoShows	Number of earlier no-shows of the same patient
PatientHistory	First appointment, Attended before or Previous no-show

`PriorNoShows` compares each row with the row above: the table is sorted by patient, then date, then appointment number, and the formula adds the no-shows of the patient's earlier rows. `PatientHistory` follows from it.

4.3 Checks

Live formulas on the clean table confirm the result: 106,987 raw rows minus 10 removed rows equal the 106,977 rows of the table; no cell is empty; no lead time is negative; the oldest patient is 102 years old; and 60,268 rows are a first appointment, which is the number of patients.

Data cleaning and preparation

The raw file holds 106,987 rows and 15 columns. 10 rows are removed; every other problem is corrected without losing a row.

1. Audit of the raw file

Check	Rows	Decision
Empty cells	0	None found: no action
Duplicate AppointmentID	0	None found: no action
Date.diff different from AppointmentDay minus ScheduledDay	0	None found: the lead time is recomputed by formula
Appointment dated before its booking (Date.diff < 0)	5	Removed: impossible chronology
Age of 115 years	5	Removed: implausible age
Patientid with decimals	5	Kept, stored as text: only the Identifier is corrupted
Second or later booking of a patient on the same day	8,532	Kept: each booking has its own AppointmentID and outcome
Appointments on a Saturday	39	Kept: rare but valid
Misspelled column names (Hypertension, Handcap, Date.diff)	3	Renamed: see table 3
Codes TRUE/FALSE, F/M and upper-case names	106,987	Recoded: see table 3

2. The 10 rows removed (as in the raw file)

AppointmentID	PatientId	ScheduledDay	AppointmentDay	Date.diff	Age	Reason
5679978	7839272661752	10/05/2016	09/05/2016	-1	38	Appointment dated before its booking
5715660	7896293967868	18/05/2016	17/05/2016	-1	19	Appointment dated before its booking
5700278	31963211613981	16/05/2016	19/05/2016	3	115	Age of 115 years
5700279	31963211613981	16/05/2016	19/05/2016	3	115	Age of 115 years
5664962	2425225838979	05/05/2016	04/05/2016	-1	22	Appointment dated before its booking
5542812	31963211613981	08/04/2016	16/05/2016	38	115	Age of 115 years
5686628	998231581612122	11/05/2016	05/05/2016	-6	81	Appointment dated before its booking
5655637	3787481966821	04/05/2016	03/05/2016	-1	7	Appointment dated before its booking
5744037	31963211613981	30/05/2016	30/05/2016	0	115	Age of 115 years
5717451	748234579244724	19/05/2016	03/06/2016	15	115	Age of 115 years

6. Checks on the clean table (live formulas)

Check	Value
Rows in the raw file	106,987
Rows removed	10
Rows in the clean table	106,977
Raw rows minus removed rows minus clean rows	0
Patients (rows with PatientHistory + First appointment)	60,268
Empty cells	0
Negative lead times	0
Oldest patient (years)	102

Expected: difference 0, empty cells 0, negative lead times 0.

Figure 1: Data Cleaning sheet: audit of the raw file, the ten rows removed, the recoding table and, on the right, the live checks on the clean table.

5 KPIs

Each KPI answers one operational question. The table gives its value on the clean table and the Excel formula that computes it on the *KPI Definitions* sheet.

Table 3: Key performance indicators

KPI	Question answered	Value
Appointments	How much demand is there?	106,977
Missed appointments	How many slots are lost?	21,672
No-show rate	How often is a slot wasted?	20.3%
Average lead time	How far ahead are patients booked?	10.2 days
Same-day share	How much activity is booked on the day?	34.7%
SMS coverage	How many patients are reminded?	32.3%
No-show rate, booked ahead	What is the risk once a patient waits?	28.5%
Repeat no-show rate	Does past behaviour predict the next visit?	29.4%
Attendance rate	How reliable is the schedule?	79.7%

The last two rate KPIs name their own base: the booked-ahead rate is computed on the 69,824 appointments booked one day or more ahead, the repeat rate on the 15,563 appointments of patients who had already missed one.

6 Results

The charts of this section are those of the *Dashboard* sheet, with every filter set to All.

6.1 Lead time

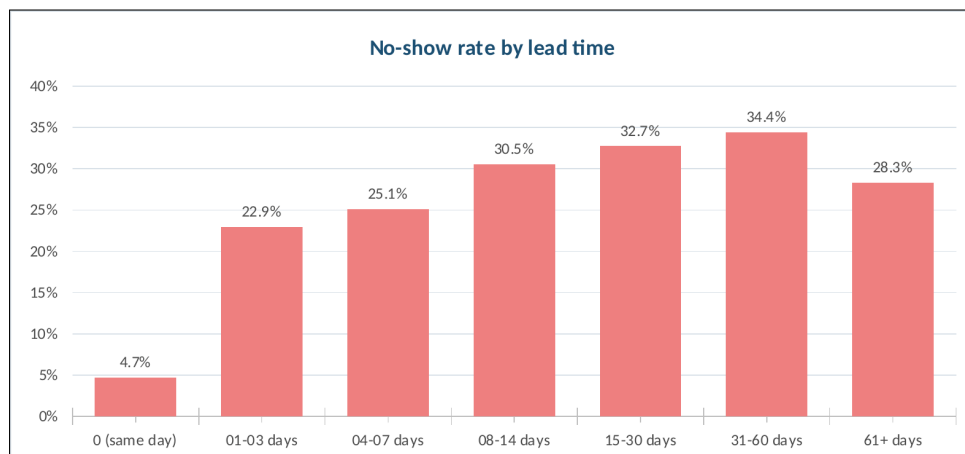


Figure 2: No-show rate by lead time. Same-day appointments are 34.7% of the volume but only 8.0% of the no-shows.

As soon as the patient has to wait, the rate jumps from 4.7% to 22.9% for 1 to 3 days and climbs to 34.4% for 31 to 60 days. The slight fall beyond 60 days concerns only 2,044 appointments.

6.2 The SMS reminder

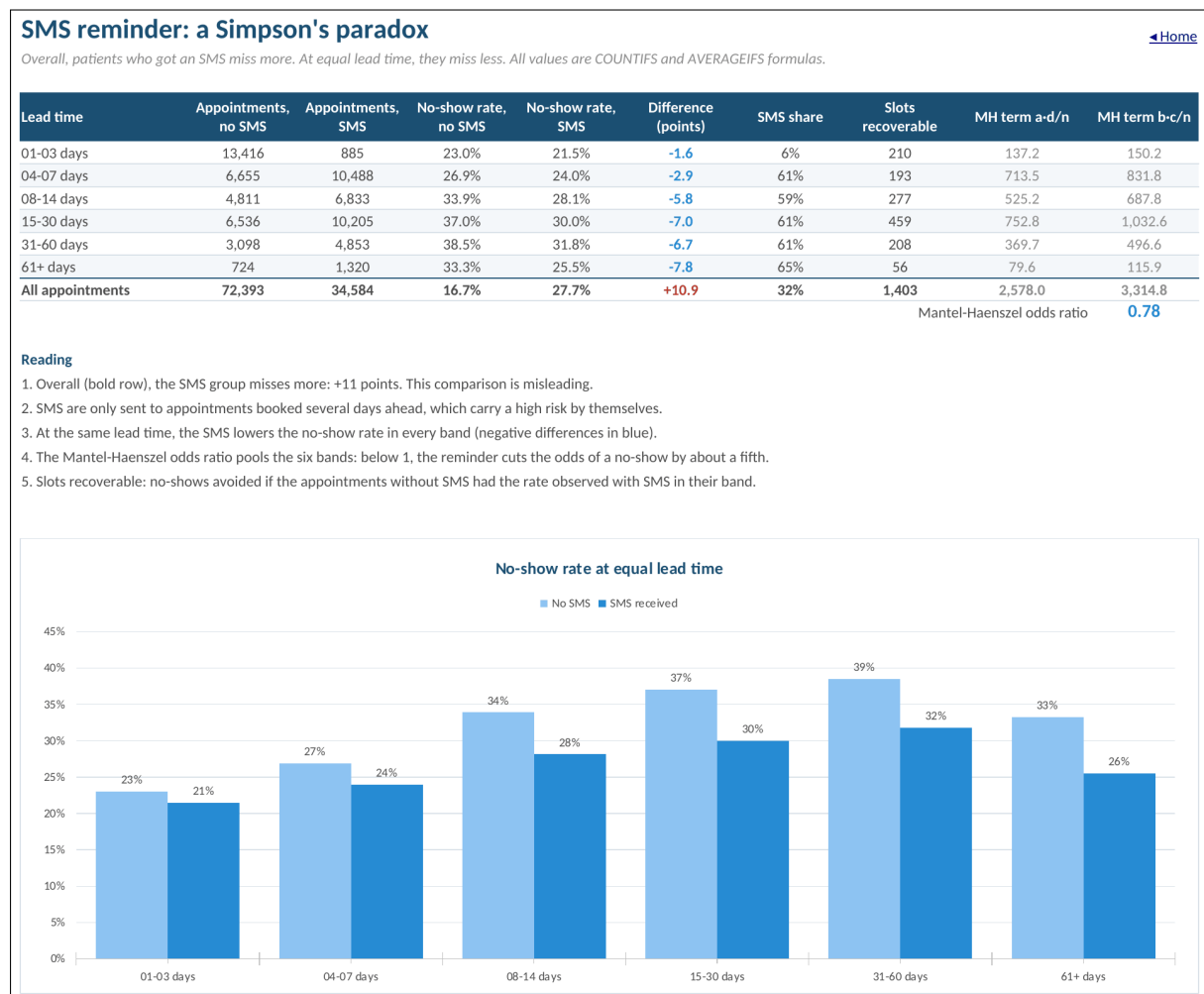


Figure 3: SMS Analysis sheet: no-show rate with and without SMS in each lead-time band (same day excluded, since no SMS is sent), Mantel-Haenszel odds ratio and slots recoverable.

Overall, the SMS group misses 10.9 points more. This is a Simpson's paradox. No SMS is sent for same-day appointments, and only 6.2% of the 1 to 3 day appointments receive one, against about 60% beyond three days: the SMS group is made of long-lead appointments, which carry the highest risk. Compared like with like, the SMS lowers the rate in every band, by 1.6 points (1 to 3 days) to 7.8 points (61+ days). The Mantel-Haenszel odds ratio, which pools the six bands, is **0.78**: at equal lead time the reminder cuts the odds of a no-show by about a fifth.

6.3 Who misses appointments

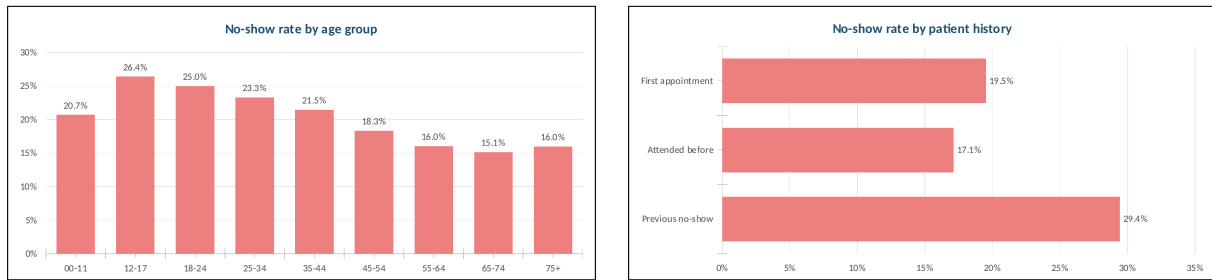


Figure 4: No-show rate by age group (left) and by patient history (right). Teenagers of 12 to 17 years miss 26.4% of their appointments, patients of 65 to 74 years 15.1%. A patient who already missed an appointment misses the next one in 29.4% of cases, against 17.1% after an attended visit.

6.4 Health and social factors

Table 4: No-show rate with and without each factor (chi-square test, 1 degree of freedom)

Factor	Group size	Rate	Others	Difference	p-value
Scholarship (welfare)	10,809	23.8%	19.9%	+3.9	< 0.001
Hypertension	21,800	17.3%	21.0%	-3.7	< 0.001
Diabetes	7,943	18.0%	20.4%	-2.4	< 0.001
Handicap	2,234	18.0%	20.3%	-2.3	0.007
Alcoholism	3,360	20.1%	20.3%	-0.1	0.872
Gender (female)	70,110	20.4%	20.1%	+0.3	0.293

Welfare recipients miss 3.9 points more. Gender and alcoholism make no difference. Hypertension and diabetes seem protective, but hypertensive patients are mostly older, and older patients miss less. Compared at equal age, the gap disappears: from 45 years, where most hypertensive patients are, their rate is equal to or higher than that of the others.

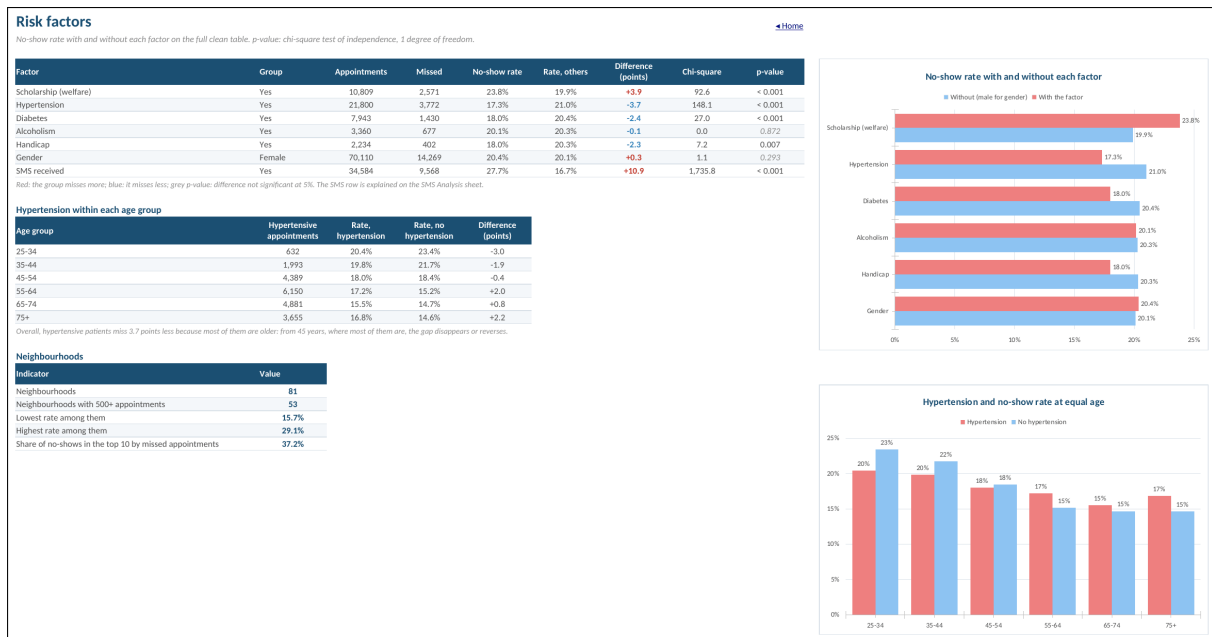


Figure 5: Risk Factors sheet: rates with and without each factor with the chi-square test, hypertension at equal age, and the neighbourhood indicators.

6.5 When and where

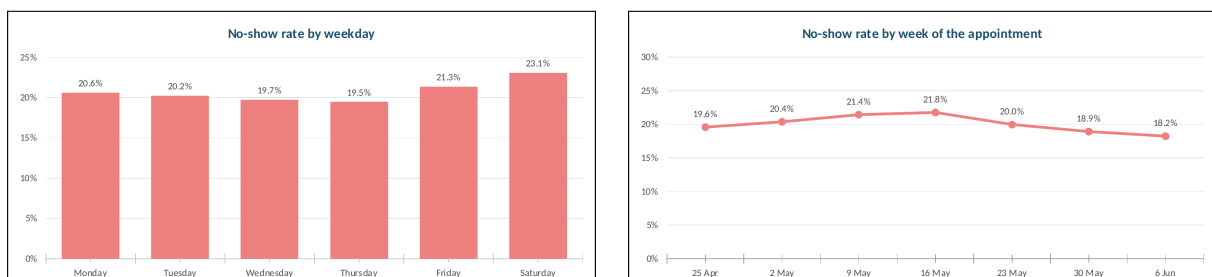


Figure 6: No-show rate by weekday (left) and by week of the appointment (right). From Monday to Friday the rate moves between 19.5% and 21.3% (the 39 Saturday appointments are too few to count); over the six weeks it stays between 18% and 22%.

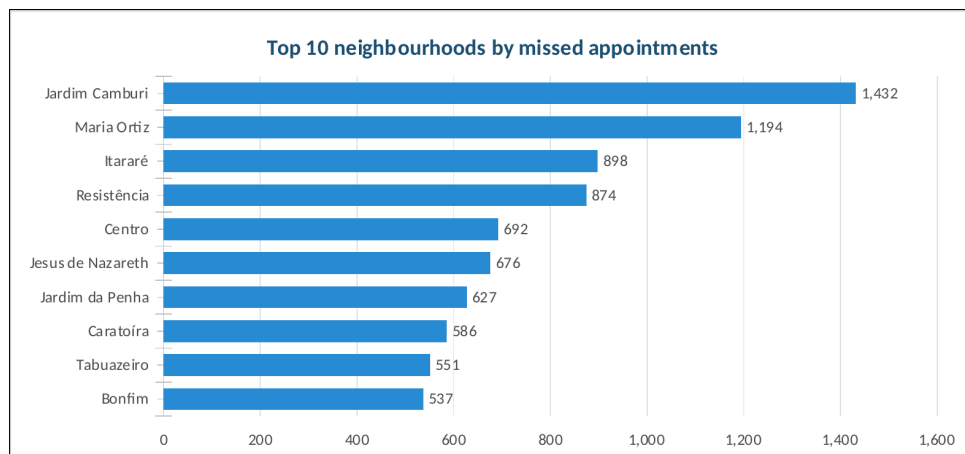


Figure 7: The ten neighbourhoods with the most missed appointments. Together they account for 37.2% of all no-shows.

Place matters more than time. Among the 53 neighbourhoods with at least 500 appointments, the no-show rate ranges from 15.7% to 29.1%, a spread of thirteen points, against less than two points between weekdays.

7 Insights and recommendations

7.1 Lead time is the primary risk

4.7% of same-day appointments are missed, against 28.5% of the appointments booked one day or more ahead.

Recommendation. Keep a share of same-day and short-notice slots, and apply a moderate overbooking to the slots booked more than a week ahead, where nearly one appointment in three is missed. *Owner: scheduling.*

7.2 The reminder works but misses its target

At equal lead time the SMS lowers the odds of a no-show by about a fifth. Yet 35,240 appointments booked one day or more ahead received no SMS, and 29.4% of them were missed.

Recommendation. Send the reminder to every appointment booked one day or more ahead, starting with the 1 to 3 day bookings, of which only 6.2% are reminded today. If these appointments had the no-show rate observed with an SMS in their band, about **1,400 slots** would have been saved over the six weeks. *Owner: patient communication.*

7.3 Past no-shows repeat

A patient who already missed an appointment misses the next one in 29.4% of cases, against 17.1% after an attended visit.

Recommendation. Flag these patients at booking and call them the day before the appointment.
Owner: front desk.

7.4 Young patients and welfare recipients need a different contact

Teenagers of 12 to 17 years miss 26.4% of their appointments and young adults of 18 to 24 years 25.0%. Welfare recipients miss 3.9 points more than other patients.

Recommendation. Address the reminders for minors to their parents and offer slots outside school hours. Review access barriers (transport, working hours) with the social services.
Owner: scheduling and social services.

7.5 Chronic patients are not safer

The lower rate of hypertensive patients is an age effect, not a behaviour.

Recommendation. Do not reduce the reminders sent to chronic patients. *Owner: care coordination.*

7.6 Place, not the weekday, is the lever

The weekday moves the rate by less than two points, the neighbourhood by thirteen, and ten neighbourhoods hold 37% of the no-shows.

Recommendation. Keep the weekly schedule unchanged and focus outreach on the ten neighbourhoods that lose the most slots. *Owner: public health.*

Key insights and recommendations					Home
Each insight is backed by a live value computed with a formula on tbAppointments (full clean table).					
#	Insight	Evidence	Live value	Recommended action	Owner
1	Lead time is the main driver of no-shows	Same-day appointments: 4.7% missed; booked one day or more ahead: 28.5%, six times more	28.5%	Keep a share of same-day and short-notice slots; overbook moderately the slots booked more than a week ahead	Scheduling
2	The SMS reminder works, but reaches the wrong appointments	27.7% with SMS vs 16.7% without overall, yet lower with SMS in every lead-time band; Mantel-Haenszel odds ratio 0.78	0.78	Send the reminder to every appointment booked one day or more ahead (only 6% of 1 to 3 day bookings get one)	Patient communication
3	Past no-shows repeat	29.4% after a previous no-show against 17.1% after an attended visit	29.4%	Flag these patients at booking and call them the day before	Front desk
4	Teenagers and young adults miss most	26.4% at 12 to 17 and 25.0% at 18 to 24 against 15.1% at 65 to 74	26.4%	Remind parents; offer slots outside school hours	Scheduling
5	Welfare recipients miss more	23.8% with scholarship against 19.9% without (chi-square $p < 0.001$)	23.8%	Review access barriers (transport, working hours) with the social services	Social services
6	Chronic conditions are not protective	Hypertension 3.7 points lower overall, but no lower within the age groups from 45 years: an age effect	17.3%	Do not reduce reminders for chronic patients	Care coordination
7	Place matters more than the weekday	Monday to Friday 19.5% to 21.3%; neighbourhoods with 500+ appointments 15.7% to 29.1%	53	Target outreach on the ten neighbourhoods that lose the most slots	Public health

Live value: row 1 booked-ahead rate; row 2 Mantel-Haenszel odds ratio (SMS Analysis); rows 3 to 6 rate of the group named; row 7 neighbourhoods with 500+ appointments (Risk Factors).

Figure 8: Insights sheet: each insight with its evidence, a live value computed on the clean table, the recommended action and its owner.

8 The Excel dashboard

`Healthcare_NoShows_Dashboard.xlsx` opens on a *Home* sheet whose contents link to every other sheet.

Table 5: Sheets of the workbook

Sheet	Content	Excel tools
Home	Contents with a link to each sheet	Hyperlinks
Dashboard	6 drop-down filters, 6 KPI cards, 8 charts	Data validation, COUNTIFS, SUMIFS
Pivot Dashboard	6 slicers, 6 KPI cards, 8 pivot charts	Slicers, pivot charts, GETPIVOTDATA
Insights	Seven insights with evidence, live value, action and owner	AVERAGEIFS
Risk Factors	Factors with chi-square tests, hypertension at equal age, neighbourhoods	COUNTIFS, CHISQ.DIST.RT, MINIFS, data bars
SMS Analysis	SMS at equal lead time, Mantel-Haenszel odds ratio	COUNTIFS, conditional formatting
Dashboard Data	Formula tables behind the Dashboard, dynamic top 10	INDEX, MATCH, LARGE
Pivot Tables	The pivot tables behind the Pivot Dashboard	Pivot tables, top 10 filter
Data Cleaning	Audit, rows removed, recoding, formulas, checks	PROPER, SUBSTITUTE, COUNTBLANK
KPI Definitions	Question, definition, formula and value of each KPI	Structured references
Clean Data	106,977 appointments, 23 columns	Excel table

The workbook offers two ways to filter. On the *Dashboard*, six drop-down lists (gender, age group, lead time, SMS received, scholarship, weekday) drive COUNTIFS and SUMIFS formulas, so that the sheet works in any spreadsheet application. On the *Pivot Dashboard*, the same six criteria are slicers connected to every pivot table: one click filters the KPI cards and the pivot charts at once. Each chart is anchored to a fixed block of cells, so it keeps its place and size whatever the screen or the zoom.

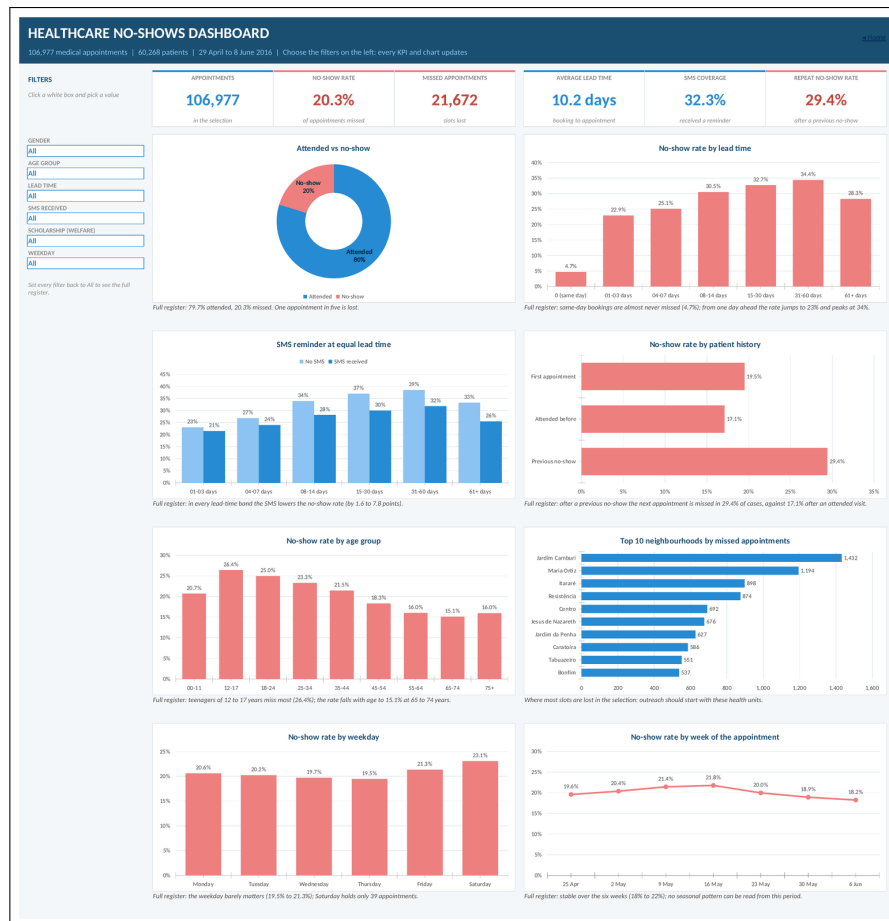


Figure 9: Dashboard sheet: six KPI cards and eight charts, all filters set to All.

9 Limitations

Six weeks of data. The file covers 29 April to 8 June 2016; no seasonality or trend can be read.

The reason for a no-show is not recorded, nor the specialty or the time of day.

The SMS flag says that a reminder was sent, not that it was read. The SMS effect is an observational estimate; a trial on the 1 to 3 day bookings would confirm it before a full roll-out.

Association, not causation. The comparisons identify the groups at risk; they do not predict an individual no-show.

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